KANPRI TECHNOLOGIES LLP

Digital MV Customer Service System

Plain-language end-to-end operational workflow, field engineer journeys, approvals and management reporting. Version 1.0 | 18 October 2026

DOCUMENT	PREPARED FOR
WORKFLOW DOCUMENT	Alfanar Electrical Systems - MV Customer Service

What this document explains

How a customer complaint becomes an assigned service job, how engineers document the site visit, how supervisors and quality teams act on it, and how the case reaches an auditable closure.

One-line workflow

Customer request → Case review → Work order → Engineer site visit → Technical inspection → Quality/material action (if needed) → Supervisor approval → Customer closure → Dashboard and reports

Who uses the system

USER	WHAT THEY DO
Public customer / contractor	Submit a new complaint and receive a reference; cannot browse existing cases.
Dispatcher / supervisor	Check intake, assign jobs, manage follow-ups and review submitted work.
Field engineer	Open assigned jobs; record travel, inspection, photos, equipment observations and actions.
QCA / store staff	Make quality decisions and record separate material receipt, issue, return and disposal actions.
Manager / auditor / administrator	Review dashboards, reports, audit records, access and approved settings within assigned permissions.

Important: The supplied interface images are concepts, not proof of working functionality. The client’s final v4 handover controls the operational behavior.

1. Main customer-to-closure workflow

01

Public complaint submission
Who: Customer / contractor
Customer enters contact, customer/site details, panel reference if known, complaint and optional attachments on a public link.
Result: A new unverified intake reference is created; the public user sees only a confirmation.

02

Review and classify complaint
Who: Dispatcher
Validate details, identify duplicates, assign priority and category; resolve site and equipment references.
Result: Case is accepted for action or returned for clarification. Original complaint remains preserved.

03

Create and assign work order
Who: Dispatcher / supervisor
Set engineer, site, target date, priority, contact and job instructions.
Result: Engineer receives an assigned work order; reassignment is tracked.

04

Engineer accepts and travels to site
Who: Engineer
Open the assigned case on a mobile-friendly screen, review equipment history, begin the first travel timer.
Result: A site-job time log is created with recorded start/end events.

05

Perform and document site work
Who: Engineer
Capture panel identification, energization, inspections, findings, measurements, photos, GPS and actions for each equipment item.
Result: An editable draft visit exists, with one site attendance and potentially multiple panel entries.

06

Record quality or material needs
Who: Engineer, QCA, store
Create defect lines, NCR, parts requests and any shutdown/customer/factory follow-up where necessary.
Result: Related actions are tracked in separate approval and custody workflows.

07

Review submitted visit
Who: Supervisor
Approve submitted revision or return with comments and requested corrections.
Result: The approved revision is preserved; no self-approval.

08

Customer closure and reporting
Who: Authorized supervisor / manager
Resolve outstanding actions, obtain appropriate confirmation, close case and generate approved report.
Result: Closed service history, searchable dashboard and exportable PDF/Excel documents.

2. Engineer workflow: a single site job

Each engineer and supervisor must have an individual staff account. No shared team login. A job is linked to a case, work order, engineer and site.

STEP	ENGINEER BUTTON / ACTION	SYSTEM RECORDS
1	Start Travel to Site	Room/base departure timestamp, engineer, site and job ID
2	End Travel at Site	Site arrival timestamp; outbound travel duration
3	Start Work	On-site work start timestamp
4	End Work	On-site work end timestamp and duration
5	Start Return Travel	Return journey start timestamp
6	End Travel at Room/base	Return end timestamp; return duration and total job time

Strict time-log rules

• Exactly three segments: Room/base-to-site travel, on-site work, and site-to-Room/base travel.

• Do not add break, waiting, lunch, sick leave or vacation as job-time segments. Those belong to a separate HR scope if requested.

• Show only the next valid button. Reject missing, repeated, reversed, negative-duration or overlapping events.

• For a second site, create a new three-segment log. For several panels at one site, use one travel/work log and several panel details.

• Capture GPS only when an authorized user taps the location action, with permission. No continuous employee location tracking.

• Supervisor may review and return a log for correction; changes retain the original event and reason.

Engineer mobile-screen checklist

SCREEN	MAIN FIELDS / ACTIONS
My assigned work	Work order, priority, scheduled date, site, customer and previous history.
Time log	Start/end the three timed segments and see duration totals.
Visit header	Arrival date, engineer, customer contact, site, GPS, limits on access or testing.
Panel details (repeat)	Voltage/family, subtype, serial, location, RTU/controller, observed energized/de-energized state.
Inspection (repeat)	Checklist result, measurements, failed/not checked reason, findings and root cause.
Action and evidence	Solution, corrective action, repeatable material lines, multiple private photos and captions.
Review and submit	Required-field warnings, open follow-up actions, submission declaration and supervisor review.

3. Technical inspection and equipment history

Equipment identification

The engineer selects family and voltage together, including 13.8/33 kV combinations for LBS/SLBS, Auto-Recloser, RMU, MRMU and SRMU, or an authorized Other value. The database stores voltage and family separately.

Additional data includes make, model, serial number, subtype, feeder/pole, site, related case, sales order and RTU/controller where relevant. Existing equipment history should appear before the visit is submitted.

Energy state and inspection controls

• Panel status when attended: Energized or De-energized; Not verified is an exception requiring reason and supervisor review.

• Electrical energy observation, work-order status, shutdown request and safety-permit approval are separate facts. The application does not authorize electrical switching.

• Inspection outcomes: Pass, Fail, Not Checked or Not Applicable; any Fail or Not Checked must have reasons and actions / approved exceptions.

• Keep the original customer complaint separate from controlled fault tags, observations, verified cause, solution and actual work.

• Each panel may have its own inspection, fault, photos, serial-number history and replacement components.

Example

SITE VISIT	PANEL RECORDS
Site A, Engineer 1, 09:00–13:00	Panel 1: 33 kV RMU / inspection / photo set
One shared visit and time log	Panel 2: 13.8 kV Auto-Recloser / fault and material request
No duplicate travel hours	Panel 3: 33 kV LBS / completed technical checklist

4. Quality, materials and conditional paths

Not every complaint goes straight to closure. The workflow branches only when the engineer identifies a defect, missing material, shutdown need, customer action or factory repair.

CONDITION	NEXT PROCESS	WHO DECIDES
Defective / replaced material	Add repeatable component lines; create NCR if appropriate; photograph, track serial and warranty.	Engineer reports, QCA reviews
Parts are required	Material request, approval, reservation / issue, installation and auditable stock ledger.	Storekeeper / authorized approver
Customer or planned shutdown required	Record request, schedule, approvals, permit reference and execution evidence separately.	Authorized operations / supervisor
Factory action required	Factory-return case, custody/repair/return status and follow-up owner.	QCA / store / supervisor
Work unresolved	Open follow-up owner, date and waiting reason; do not falsely close case.	Supervisor
Work completed	Submit technical report revision; supervisor approves; customer closure when applicable.	Supervisor / customer as defined

Approval principle

Separate decisions remain separate: engineer submission, supervisor technical approval, QCA NCR decision, store material movement, shutdown authorization and final case closure. One event must not silently approve another.

Revision and audit principle

Submitted records are preserved. If a supervisor returns a report, the engineer creates a corrected revision. The system records who changed what, when and why, and prevents engineers from approving their own closures.

5. Manager dashboard and reporting

DASHBOARD AREA	WHAT A MANAGER CAN SEE
Live overview	New, assigned, in progress, waiting, overdue, review pending, resolved and closed case counts.
Operations queue	Assigned engineer, site, equipment, priority, next action and due date.
Technical history	Serial-number-level equipment records, visits, failures, measures, materials and documents.
Quality / material	Open NCRs, QCA decisions, material requests, returns and custody status.
Timesheet review	Time by date, engineer, site, job and segment; incomplete logs; corrected/approved records.
Filters and export	Date, customer, region, panel, serial, category, status and owner; PDF/Excel downloads.
Audit and oversight	History of assignments, revisions, decisions and exports.

Example management questions the system answers

- How many complaints are still pending or overdue, and who owns the next action?
- Which panel serial number has repeated faults or previous replacement history?
- How long did an engineer spend travelling and working at each site?
- Which jobs are waiting on parts, customer approval, shutdown or factory return?
- Which visit reports were approved, returned or revised, and by whom?

Records and privacy

All production data belongs in the application database and private file storage, not in spreadsheets used as live data. Public submitters never see dashboards, records, photos, maps, or report exports. Backup/restore must be demonstrated before handover.

6. End-to-end delivery and sign-off

RELEASE	WHAT THE CLIENT WILL TEST
Design sign-off	Approved mobile/desktop screens, field options, checklist rules, access rules and report samples.
Core operational release	Public complaint, dispatcher assignment, engineer job timer, visits, multiple panels, photo upload, supervisor approval and dashboard.
Quality and stock release	NCR/QCA, defective material rows, requests, stock movements, asset history and customer closure.
Specialist release	Standard shutdown, remote service, commissioning, maintenance, training, mapping and reports.
Final go-live	Role-based access, audit, backup restore, reports, system performance, client-owned deployment and handover.

Client decisions required before the build

- Approve final user roles, site/asset catalogues, checklists, fault categories and report templates.
- Resolve the documented 22-form catalogue versus the separately defined F23 Staff Work Time Log.
- Confirm what is included in the agreed 10–12 week version and which advanced requirements are change requests.
- Approve hosting region, security/identity policy, retention policy and data migration approach.

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